Welcome!

Lesson 1

What is Programming?

process of writing instructions (code) to tell a computer how to solve a problem

Why Python?

it is very useful to learn python because it can be used for web development, data visualization and machine learning.

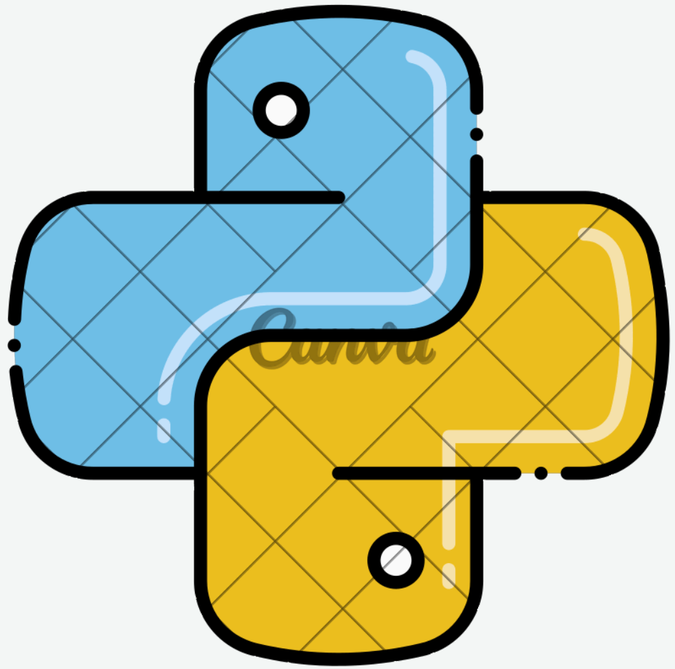
premier language for Artificial Intelligence, Data Science, and Automation

What are we learning today?

Variables

Data Types

User Input



Python/Code Editor Installed?

1) Go to python.org
2) Click Downloads and download the latest version of python
3) Go to the website for VISUAL STUDIO CODE
4) Download the one for your device

Basics

Variables:

- containers for storing values

naming rules-

- must start with a letter or the underscore character
- cannot start with a number
- can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- case sensitive (hey vs HEY)
- cannot be one of the python keywords

Valid names-

x = 3
var = “sunny”
var_check = ‘correct’

Invalid names-

1x = 3
var# = “sunny”
var_check = ‘correct’

*Syntax Error

Keywords

and	as	assert	async	await
break	class	continue	def	del
elif	else	except	False	finally
for	from	global	if	import
in	is	lambda	None	nonlocal
not	or	pass	raise	return
True	try	while	with	yield

Cannot name variables with one of these

Data Types

Boolean Type: **True/False**

1) set boolean value →

check_guess = True
has_permission = False

2) generate boolean value →

print(10>5) → True

None Type:

-no value assigned yet
-placeholder

-defines the classification of a data value

numeric:

integer, float

int: 1,2,0,-3
float: 3.45, 2.6, 5.0

strings:

-immutable sequence of characters to define text data
-created by enclosing characters inside single, double, or triple quotes

str = “Hello World!”

Quizziz!

Over variables and data types

Thank You!

Any Questions?